Differentially Private Synthetic Data

Version: 2

Differentially Private Synthetic Data

- Read the below article and apply the *differential privacy* scheme on your HW1 result and dataset, replacing the methods of *k-anonymity, and ℓ-diversity*.

  Differentially Private Synthetic Data | NIST

  - The winners' codes is listed in the next page. You can choose one or several to make the experiments.

- In your document report, you should provide:
  - Describe your method (from the webpages and their academic papers)
  - **Describe your experimental results**
  - Provide some novel ideas or schemes in this project (if you have)
  - **Provide a comparison with HW1**, e.g., both released the obfuscated dataset.
  - **Provide a comparison with HW2**, e.g., differential privacy is applied at different stages.

Reference (Code)

- De-identification Tools | NIST
  - **Final Challenge Winners:**
    1st Place - Team RMcKenna
    2nd Place - Team DPSyn
    3rd Place - Team PrivBayes
    4th Place - Team Gardn999
    5th Place - Team UCLANESL

Upload the result to Moodle

- Upload your code and the ***detailed*** reports to Moodle.
  - Code: You must add some comments in your code for easy understanding.
  - Report Files: Document (Word) and Presentation (PPT).
    - Including a table in your report outlining the responsibilities of each team member for this homework.
  - Video Record: Web address of oral presentation uploaded
  - Zipped to a file. The file name should be “TeamName_Midterm.zip”